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Discussion Agenda 05/06/2015 (Isolated Disk + AGORA Paper 4)

Below is a discussion agenda / live progress report for the 7th Web Conference of the AGORA Project (May 6, 9 am PDT/6 pm CEST). This telecon will particularly focus on the progress and future work for AGORA Paper 4, with some updates from Paper 3. But any interested member is welcome to join. Some items below were discussed among the Steering Committee members and Paper Group leaders on Apr. 15.

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[1. Progress Report \(as of May 2015\)](#)

We are actively making a lot of progress, chiefly, in three different fronts below (numbered tentatively).

	PAPER "2"	PAPER "3"	PAPER "4"																																																														
GOALS	• Run Proof-of-concept-DM (~1e11 Ms halo)	• Implement and test common astrophysics package • Run Proof-of-concept-NOSUBGRID (~1e11 Ms halo)	• Calibrate SF/Fbck recipes using common disk ICs																																																														
LEADERS	• Oliver Hahn (OCA, Nice)	• Nick Gnedin (Fermilab) • Piero Madau (UCSC) • Britton Smith (Edinburgh)	• Oscar Agertz (Surrey) • Romain Teyssier (Zurich) • James Wadsley (McMaster)																																																														
PARTICIPANTS	• Working Groups engaged: – III (common cosmo ICs) – IV (common analysis) – VI (dwarf galaxies) • Code representatives: <table><tr><th>Code</th><th>Contacts</th></tr><tr><td>ART-I</td><td>Klypin</td></tr><tr><td>ART-II</td><td>Gnedin, Leitner</td></tr><tr><td>CHANGA</td><td>(Quinn)</td></tr><tr><td>ENZO</td><td>Kim, Simpson, Wise</td></tr><tr><td>GADGET</td><td>Choi, Hahn, Nagamine, Onorbe, Rocha, Todoroki</td></tr><tr><td>GADGET-SPHS</td><td>Hobbs</td></tr><tr><td>GASOLINE</td><td>Shen, Zolotov</td></tr><tr><td>GCD+</td><td>Kawata</td></tr><tr><td>PKDGRAV2</td><td>Guedes, (TBA)</td></tr><tr><td>RAMSES</td><td>Hahn, Teyssier</td></tr></table>	Code	Contacts	ART-I	Klypin	ART-II	Gnedin, Leitner	CHANGA	(Quinn)	ENZO	Kim, Simpson, Wise	GADGET	Choi, Hahn, Nagamine, Onorbe, Rocha, Todoroki	GADGET-SPHS	Hobbs	GASOLINE	Shen, Zolotov	GCD+	Kawata	PKDGRAV2	Guedes, (TBA)	RAMSES	Hahn, Teyssier	• Working Groups engaged: – I (common physics) – III (common cosmo ICs) • Code representatives: <table><tr><th>Code</th><th>Contacts</th></tr><tr><td>ART-I</td><td>Ceverino, Klypin</td></tr><tr><td>ART-II</td><td>Gnedin, Letiner</td></tr><tr><td>CHANGA</td><td>Quinn</td></tr><tr><td>ENZO</td><td>Hummels, Kim, Simpson, Smith, Turk, Wise</td></tr><tr><td>GADGET</td><td>Choi, Nagamine</td></tr><tr><td>GADGET-SPHS</td><td>Hobbs</td></tr><tr><td>GASOLINE</td><td>Madau, Mayer, Shen, Wadsley</td></tr><tr><td>GEAR</td><td>Revaz</td></tr><tr><td>GIZMO</td><td>Lupi, Thompson</td></tr><tr><td>RAMSES</td><td>Agertz, Butler, Teyssier</td></tr></table> (Grackle implementers, identified as of May 6, 2015)	Code	Contacts	ART-I	Ceverino, Klypin	ART-II	Gnedin, Letiner	CHANGA	Quinn	ENZO	Hummels, Kim, Simpson, Smith, Turk, Wise	GADGET	Choi, Nagamine	GADGET-SPHS	Hobbs	GASOLINE	Madau, Mayer, Shen, Wadsley	GEAR	Revaz	GIZMO	Lupi, Thompson	RAMSES	Agertz, Butler, Teyssier	• Working Groups engaged: – II (common disk ICs) – IV (common analysis) – V (subgrid physics) • Code representatives: <table><tr><th>Code</th><th>Contacts</th></tr><tr><td>ART-I</td><td>Arraki, Ceverino, Klypin, Mandelker, Snyder</td></tr><tr><td>ART-II</td><td>Feldmann, Gnedin, Leitner</td></tr><tr><td>CHANGA</td><td>Quinn</td></tr><tr><td>ENZO</td><td>Abel, Butler, Butsky, Forbes, Hummels, Kim, Simpson, Turk, Wise</td></tr><tr><td>GADGET</td><td>Choi, Nagamine, Thompson, Todoroki</td></tr><tr><td>GADGET-SPHS</td><td>Hobbs</td></tr><tr><td>GASOLINE</td><td>Benincasa, Keller, Madau, Mayer, Shen, Wadsley</td></tr><tr><td>GEAR</td><td>Revaz</td></tr></table>	Code	Contacts	ART-I	Arraki, Ceverino, Klypin, Mandelker, Snyder	ART-II	Feldmann, Gnedin, Leitner	CHANGA	Quinn	ENZO	Abel, Butler, Butsky, Forbes, Hummels, Kim, Simpson, Turk, Wise	GADGET	Choi, Nagamine, Thompson, Todoroki	GADGET-SPHS	Hobbs	GASOLINE	Benincasa, Keller, Madau, Mayer, Shen, Wadsley	GEAR	Revaz
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PROGRESS MADE OR BEING MADE	• Please visit Paper 2's Workspace where Oliver Hahn et al. have been updating the progress report. • 5 out of 8 convex hull ICs (ellipsoidal ICs for cosmological halos) have been generated by several people. • Thanks to Miguel Rocha et al., a multi-mass version of Rockstar is implemented and being tested on a yt platform, as of Mar. 2015.	• Thanks to the work by Britton Smith, GRACKLE-2.0 is released on Oct. 1, 2014 with functioning APIs for Fortran, C, C++ codes, options for equilibrium and non-equilibrium coolings, and improved documentation. • Again thanks to the great effort by Britton Smith et al., GRACKLE-2.1 is released on June 2, 2015 addressing issues raised by the AGORA community, including (1) one-zone cooling test, (2) updated tabulated cooling when UV is on, and (3) updated solar metallicity value. See Section 2-(1) below for more details. • Nick Gnedin has also coded up and released an actual AGORA-GRACKLE API on Apr. 7, 2015 (not dummy; compatible with Grackle 2.0). • Either via its original interface or through Nick's API, GRACKLE is now fully implemented in 7+ codes including ART-II, ENZO, GADGET, GASOLINE, GEAR, GIZMO, and RAMSES, as of May 6, 2015. 3 remaining groups — ART-I, ChaNGa, and Gadget-SPHS — are taking advantage of the Grackle interfaces in their sibling codes — ART-II, Gasoline, and Gadget, respectively — to complete their own implementations. The resulting cooling functions will be compared and verified in isolated disk runs (Paper "4").	• Please visit Paper 4's Workspace where Oscar Agertz et al. have been updating the progress report. • 8+ participating codes including GIZMO have completed their preliminary tests and basic yt visualization, as of Apr. 2015. For example, see Section 4 of Paper 4's Workspace. • The agora-analysis-script repository will be updated as the analysis progresses. Meanwhile, the yt-SUNRISE component of the analysis is already in there thanks to the effort by Miguel Rocha, Greg Snyder, et al. • See Section 2 below for more items to discuss today and the 2015 timeline.				
OTHER LINKS	• Please see the progress report in Paper 2's Workspace and/or in WG III's Workspace.	• GRACKLE Google group • GRACKLE download & documentation	• Please see the progress report in Paper 4's Workspace and/or in WG V's Workspace.				

	• MUSIC Google group	• AGORA API for GRACKLE by Nick Gnedin	• yt-AGORA Google group
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[2. Paper Group 4 - Timeline Before 2015 Summer Workshop](#)

- Per Steering Committee's recommendation, Paper 4 ([workspace](#)) will be an area of focused AGORA efforts before this year's August workshop.
- The following items in Section 2 are here for archival purpose. Updated information can be found in the collaborative document in their [workspace](#).

(1) May 6: FIRST WEB CONFERENCE

(1-1) Discussion Item #1 – Logistics

- Identify at least one *devoted* person per each code.
[Done via individual email conversations with each groups. Check the names in blue in the table above.]
- Ratify the near-term timeline.
[Done.]
- Given the time constraint, can we narrow or redefine the scope of Paper (i.e. focus on comparison of only 2 types of simulations: (1) Grackle+no SF run, (2) Grackle+SF+thermal fbck run)?
[Agreed. But we may consider Grackle+SF+favorite fbck runs later as a subset comparison project.]
- For the same reason, may we opt to reduce the number of analysis shopping lists?
[Agreed. We will reconsider the list as we move along.]
- Grackle is now fully implemented in 7+ codes listed above in the table. Can we set a deadline for remaining code groups to either implement Grackle or identify their own Grackle-compatible cooling?
- We will use isolated disks as testbeds for Grackle, so may we consider incentivizing Grackle implementers by giving them credits in Paper 4 (not just 3)?
- Ji-hoon is willing to volunteer to collect the datasets for comparison (eventually at NERSC), and run some initial cross-platform analysis.

(1-2) Discussion Item #2 – Requirements for First Datasets (Grackle+no SF run)

- Need to agree on the specific requirements for the first datasets to be submitted.
[We agreed to the following list:
- 1) No SF +
- 2) Low-resolution (i.e. $M_{\text{gas}}=8.593 \times 10^4 M_{\text{sun}}$ or $dx=80$ pc, described in the readme file) +
- 3) Grackle cooling (tabulated primordial cooling by Cloudy at Z_{sun} + tabulated metal cooling by Cloudy + UV heating by Haardt & Madau 2012; see [Grackle doc](#) for more info) +
- 4) Refinement criteria for AMR is left to each participants to find a best way to match SPH (but we recommend refinement if over $8 * M_{\text{gas}}$ + additional refinements on DM and star particles) +
- 5) Jeans pressure floor (described in the readme file; this may not be needed after all and may be worth investigating, so if you want to experiment it without the floor please do so and submit it later)]
- (all these and more discussed in the [previous workshop](#) and documented in the dropbox [readme file](#))
- One-zone/one-particle test for Grackle cooling? (test set-up, what metric/tolerance to use to compare across codes? Yves and James may share their stories?)
[Participants are not required to submit this result, but one may want to make sure his Grackle implementation is done right. A benchmark one-zone cooling test is recently added in Grackle (outputs a t-T plot; [pull request](#) as of May 11, 2015; now [accepted](#) in Grackle 2.1 as of June 2, 2015). Another possible benchmark is in page 14/15 of [Oscar's talk](#) in last year's meeting (they report that Grackle and Ramses coolings agree with each other well). Yet another test exists in Nick's API.]
- Possible issues in Grackle with (1) mean molecular weight and (2) solar metallicity?
[– First, James and Britton come to realize that a fully tabulated cooling library as a function of only temperature may produce inconsistent results if UV heating is included. This is because the mean molecular weight (μ) currently depends on CIE temperature only when in reality UV can make low-dense medium fully ionized. Ongoing effort to update Grackle effectively makes the tabulated cooling as a function of density as well when UV is on ([link1](#) [link2](#)). At the same time, Grackle community including Britton is carefully changing the old solar metallicity value Z_{sun} in the

code [link1](#), [link2](#), [link3](#). These two changes are now **incorporated** in Grackle 2.1 as of June 2, 2015. (Notes updated May 28 and June 2, 2015)
 – Despite these planned issues, we ask you to still go ahead with the existing Grackle 2.0 and submit first datasets as planned. As long as the same standard is shared across AGORA we can still start a consistent cross-platform analysis. We may later collectively switch to the updated Grackle if we deem it necessary. (Notes added on May 11, 2015)

(1–3) Discussion Item #3 – Requirements for Second Datasets (Grackle+SF+thermal fbck run)

- Need to check if SF+thermal fbck recipes in each codes are compatible with what was described in the Flagship paper, and within SPH/AMR peers. [We agreed to the following:

- 1) Simple Schmidt-like SF recipe (described in the readme file, with free-fall time $t_{\text{ff}} = \sqrt{3\pi/(32G\rho_{\text{gas}})}$)
- 2) Thermal feedback of Type II SN only (1e51 ergs/SN, its detailed implementation and other related subgrid physics such as IMF and enrichment described in Kim et al. (2014) are left to participants' discretion)]

(all discussed in the [previous workshop](#) and documented in the dropbox [readme file](#))

- Once confirmed as compatible, participants should immediately start SF+thermal fbck runs, and submit the results.

(2) May 6 – May 31: FIRST DATASET SUBMISSION (Grackle+no SF run)

- Submit first datasets for comparison and let Ji-hoon know.

[For starters, submit only two snapshots at 0 Gyr and at 0.5 Gyr. But make sure to store at every 50 Myrs for future analysis. If your simulation is slower than expected, feel free to first submit an early snapshot, e.g. 50 Myr, so we can already start cross-platform analysis. If you don't have a user account at NERSC check [here](#)]

- Ji-hoon, Cameron, and other volunteers can run some initial cross-platform analysis at NERSC or on other platforms, and already work with yt representatives to resolve outstanding issues.

[The [agora-analysis-script](#) repository will be updated as the analysis progresses.]

- Produce multi-panel images and plots on the web, and iterate until each code group arrives at a parameter configuration that makes its code compatible with others.

[We will start with the two basic diagnostics:

- 1) a gas surface density map +
- 2) a rho–T phase plot]

- Make sure that either Grackle or Grackle-compatible cooling is reliable, and compatible across all codes.

(3) June 1 – June 30: SECOND DATASET SUBMISSION (Grackle+SF+thermal fbck run)

- Ask participants to submit second datasets (Grackle+SF+thermal fbck runs) as they become available.
- Go through another round of iteration process with each code groups.

(4) Late June: SECOND WEB CONFERENCE

- More to come: paper writing, etc.

3. Workshops This Year

- Joel and Piero want to remind you that 2015 Santa Cruz Galaxy Workshop will be August 17–21 followed by the usual weekend AGORA workshop August 21–23. See the following webpages for details.

<http://hipacc.ucsc.edu/GalaxyWorkshop2015.html>

<https://sites.google.com/site/santacruzcomparisonproject/workshops/4th-workshop>

[4. Recaps](#)

(1) yt+AGORA Pipeline on the NERSC Data Pilot Resources (PI Joel Primack):

- Please see Miguel Rocha's [talk](#) during the 3rd Workshop, and [this link](#) for more information. Thanks very much to Miguel Rocha, Alex Bogert, Matthew Turk, Joel Primack, and many others. If you represent a code group but don't have a NERSC user account, visit [this page](#). Select "standard" account and find the repository name "agora". Joel Primack will add your name to the group when approved.
- We may choose to migrate from NERSC (LBNL/DOE-site) to NCSA (UIUC/NSF-site) as we may not expect GPU support for the visualization at NERSC.

(2) Strategy to Store Simulation Outputs:

Please visit [this link](#).

(3) SMBH Accretion and Feedback Recipes:

Please visit [this link](#).

(4) Mini-workshop Opportunities:

Mini-workshop for individual paper groups to write papers: UC-HiPACC@UCSC and KIPAC@Stanford are willing to provide funding opportunities for working group leaders to organize mini-workshops (~10 people). An example was the [yt-AGORA mini-workshop](#) at UCSC in Mar. 2014.

(5) Useful Links:

- Collection of softwares relevant to AGORA effort: [here](#)
- Quick links to the important pages in AGORA Workspace: [here](#)
- Internal policy regarding the disclosure of AGORA simulation results: [here](#)

[5. Your Feedback](#)

Let us hear anything AGORA, including the scientific and organizational effort currently being made. Voice your opinion during the web conference! Feel free to share your thoughts in the comments section below, or on the [mailing list](#). You can also directly contact the [project coordinator](#) anytime!